SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE TOOLS DOCUMENT

PROJECT NAME: Happy Tails

GROUP MEMBERS: Semih Öztürk, Mirza Kerem Kılıç, Melike Gülmine Kırpat, Özge Öztürk, Hüseyin Günçiçek, Arda Özkaranfil, İdil Buse Zengin

TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	Database System
2	Software Framework
3	Live Streaming

SOFTWARE TOOLS FOR TASK 1: Database System

Tool Cost/Training/Functionality Data

Tool	MySQL	Firebase	PostgreSQL	MongoDB
Cost	$0	$25	$0	$0
Training Days	5 days	2 days	6 days	4 days
Functionality	80	95	90	85

Normalized Cost/Training/Functionality Data

Tool	MySQL	Firebase	PostgreSQL	MongoDB
Cost	0.0	100.0	0.0	0.0
Training Days	83.3	33.3	100.0	66.7
Functionality	84.2	100.0	94.7	89.5

Normalized Tool Graph

Which tool has been selected? Why?

Firebase is selected because it offers the most comprehensive features for our shelter management system, particularly for real-time updates and notifications. While it has a cost of $25, its high functionality score and short learning curve (2 days/33.3 normalized) make it the most efficient choice for rapid development.

SOFTWARE TOOLS FOR TASK 2: Software Framework

Tool Cost/Training/Functionality Data

Tool	Flutter	React Native	.NET MAUI	Ionic
Cost	$0	$0	$0	$0
Training Days	15	10	20	5
Functionality	95	85	90	70

Normalized Cost/Training/Functionality Data

Tool	Flutter	React Native	.NET MAUI	Ionic
Cost	100.0	100.0	100.0	100.0
Training Days	75.0	50.0	100.0	25.0
Functionality	100.0	89.5	94.7	73.7

Normalized Tool Graph

Which tool has been selected? Why?

Flutter was selected because its single codebase ensures high performance and consistent UI across all platforms. Additionally, its Hot Reload feature significantly speeds up development 7and reduces long-term maintenance costs.

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SOFTWARE TOOLS FOR TASK 3: Live Streaming

Tool Cost/Training/Functionality Data

Tool	WebRTC	Agora	AWS IVS	YouTube Live API
Cost (per 1k min)	$0	$0.59	$2	$0
Training Days	15	5	8	3
Functionality	85	95	90	75

Normalized Cost/Training/Functionality Data

Tool	WebRTC	Agora	AWS IVS	YouTube Live API
Cost	0.0	29.5	100.0	0.0
Training Days	100.0	33.3	53.3	20.0
Functionality	89.5	100.0	94.7	78.9

Normalized Tool Graph

Which tool has been selected? Why?

Agora was selected for providing the best balance between cost and performance. While it has a cost of $0.59 per 1k minutes, it delivers a high functionality score for low-latency streaming. Additionally, its short training period of 5 days allows our team to implement the live feed feature rapidly, ensuring we stay on schedule for other critical development tasks.